

First-Principles Derivation of the Steiner Sentence Length Distribution

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Abstract

A *Steiner word* (Steiner circuit [1]) is a path segment of an odd Collatz orbit whose compressed mod-8 form matches $(7^*3)?(1 \mid 5)$: a sequence of Syracuse-map applications beginning at any letter in $\{1, 3, 5, 7\}$ and terminating at the terminal letter 1 or 5. A *Steiner sentence* is a string of $k \geq 1$ consecutive Steiner words comprising a *clause* (the first $k - 1$ words, each terminating at letter 1) followed by a *closing word* (the final word, terminating at letter 5); its *length* is the number of words k . We prove that the sentence length distribution is $P(\text{length} = k) = 3^{k-1}/4^k$ for all $k \geq 1$.

The proof models sentences as first-passage times of a Markov chain on $\{1, 3, 5, 7\} \pmod{8}$, whose transition matrix is computed row by row from elementary arithmetic. The key structural fact is an invariant $d_3 = d_7$ — equal weight on residue classes 3 and 7 in the non-terminated distribution — which holds unconditionally and forces the surviving mass to decay by exactly $3/4$ per word.

The proof requires no ergodic theory and no appeal to the Collatz conjecture. It rests on two arithmetic facts: that $v_2(3n + 1)$ is a pointwise constant for $n \equiv 1, 3, 7 \pmod{8}$ (giving exact rational transition probabilities), and that $\gcd(3, 2^j) = 1$ (giving uniform output from $n \equiv 5 \pmod{8}$, proved in the companion paper [3]). All results have been machine-verified in Lean 4/Mathlib with no `sorry`.

Appendix B provides a complete empirical validation: 10^5 sentences sampled uniformly over $[1, 10^{15}]$, with χ^2 goodness-of-fit tests. The naive $(1/2)^k$ prediction is rejected with $\chi^2 \approx 10^6$; the formula $3^{k-1}/4^k$ is consistent ($p \approx 0.61$).

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1 Background and Notation

We work throughout with odd positive integers.

Definition 1.1 (Syracuse map). The *Syracuse map* on odd positive integers is

$$S(n) = \frac{3n + 1}{2^{v_2(3n+1)}}, \quad n \text{ an odd positive integer,}$$

where $v_2(m)$ denotes the 2-adic valuation of m .

Definition 1.2 (Steiner word). A *Steiner word* (or *Steiner circuit* [1]) is a path segment of an odd Collatz orbit whose compressed mod-8 form matches the regular expression $(7^*3)^?(1 \mid 5)$: a sequence of Syracuse-map applications whose successive input residues mod 8 spell a string over the alphabet $\{1, 3, 5, 7\}$ of that form, terminating when the input residue is 1 or 5 (the two *terminal letters*). The *entry letter* of a word is the residue modulo 8 of its first node; the *terminal letter* is 1 or 5.

Definition 1.3 (Steiner sentence). A *Steiner sentence* of length $k \geq 1$ is a string of k consecutive Steiner words in a Collatz orbit comprising a *clause* (the first $k - 1$ words, each terminating at letter 1) followed by a *closing word* (the final word, terminating at letter 5).

The full Steiner language hierarchy (alphabet, letters, words, sentences, and higher-level structures) is developed in Appendix A.

Remark 1.4. A sentence of length k corresponds to a path of k edges in the overlay tree whose nodes are odd integers $\equiv 5 \pmod{8}$ and whose edges are individual Steiner words: consecutive words in the clause terminate at letter 1 and extend the current edge, while the closing word terminates at letter 5 and completes it. This paper is self-contained; the Steiner circuit decomposition, the regular-expression characterisation, and the overlay tree will be developed in full generality in future work, but are not required here.

Remark 1.5 (Why Steiner sentences are the right scale). The Steiner sentence decomposition is chosen precisely so that class $5 \pmod{8}$ is *always a sink and never a source* within a sentence. This has a crucial arithmetic consequence.

For $n \equiv 1, 3, 7 \pmod{8}$, the 2-adic valuation $v_2(3n+1)$ is a *constant* determined entirely by $n \pmod{8}$: it equals 2, 1, and 1 respectively (Paper 66 [3], Theorem 2.1). The Syracuse image $S(n) \pmod{8}$ is therefore completely determined by $n \pmod{8}$ alone, with exact rational transition probabilities.

For $n \equiv 5 \pmod{8}$, by contrast, $v_2(3n+1) = 3 + v_2(3k+2)$ where $n = 8k+5$, and $v_2(3k+2)$ is an unbounded random variable whose expectation requires summing a geometric series (Paper 66). The transition probabilities out of class 5 are still exactly $1/4$ each, but they arise from a distribution over infinitely many valuation values, not a single fixed shift.

The Markov chain on entry letters tracks how the entry letter of each word determines the entry letter of the next, via the transition matrix P . Letter 5 is the unique terminal letter that ends a sentence (the closing word); letter 1 is the other terminal letter, but it begins the next word rather than ending the sentence. Within a sentence the surviving chain operates on entry letters in $\{1, 3, 7\}$: a word with entry letter 5 would have been the closing word of the *previous* sentence, so letter 5 never acts as an entry letter within the clause. For entry letters 1, 3, and 7, the 2-adic valuation $v_2(3n+1)$ is a *constant* determined by the entry letter alone, so every transition probability is an exact rational. The variable-valuation complexity of letter 5 is absorbed at the sentence boundary and plays no role in the counting argument. This is why the sentence length distribution $3^{k-1}/4^k$ admits a clean closed-form proof: at the scale of Steiner sentences (strings of words), the arithmetic is exact.

Paper 65 [2] was the first attempt at this subject. It observed empirically that $P(\text{sentence length} = k) = \frac{1}{3}(3/4)^k$, equivalently $3^{k-1}/4^k$, and stated the result as a conjecture. The present paper supersedes Paper 65 by providing a complete proof, and incorporates its experimental record in Appendix B.

2 The Syracuse Transition Matrix mod 8

We model the sentence as a Markov chain on the *entry letter* of each successive Steiner word — the residue modulo 8 of the word's first node, a letter in the alphabet $\{1, 3, 5, 7\}$. Each word terminates at a terminal letter (1 or 5); this terminal letter is the entry letter of the *next* word (since the terminal node of one word is the starting node of the next). A sentence ends when a word terminates at letter 5 (the closing word). The transition probability $P[a][b]$ is therefore the natural density of odd integers $n \equiv a \pmod{8}$ for which $S(n) \equiv b \pmod{8}$:

$$P[a][b] = \lim_{N \rightarrow \infty} \frac{1}{N} \# \{n \leq N : n \equiv a \pmod{8}, S(n) \equiv b \pmod{8}\}.$$

All results in this paper are unconditional: they require no assumption beyond this standard density definition and do not depend on the Collatz conjecture.

Theorem 2.1 (Syracuse transition matrix mod 8). *The transition matrix on $\{1, 3, 5, 7\} \pmod{8}$ is:*

$$P = \begin{array}{c|cccc} & 1 & 3 & 5 & 7 \\ \hline 1 & 1/4 & 1/4 & 1/4 & 1/4 \\ 3 & 1/2 & 0 & 1/2 & 0 \\ 5 & 1/4 & 1/4 & 1/4 & 1/4 \\ 7 & 0 & 1/2 & 0 & 1/2 \end{array}$$

(rows = source letter, columns = target letter).

Proof. We prove each row by direct arithmetic.

Row 1 (uniform). Write $n = 8k + 1$. Then $3n + 1 = 24k + 4 = 4(6k + 1)$. Since $6k + 1$ is always odd, $v_2(3n + 1) = 2$ exactly and $S(n) = 6k + 1$. Now $6k + 1 \pmod{8}$: as k runs through $0, 1, 2, 3 \pmod{4}$, we get $6k + 1 \equiv 1, 7, 5, 3 \pmod{8}$ respectively. Each residue in $\{1, 3, 5, 7\}$ appears with equal density, so $P[1][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Row 5 (uniform). Write $n = 8k + 5$. Then $3n + 1 = 8(3k + 2)$, so $v_2(3n + 1) = 3 + v_2(3k + 2)$. Write $3k + 2 = 2^m q$ with q odd; then $S(n) = q$. We need $q \pmod{8}$ to be uniform on $\{1, 3, 5, 7\}$.

Since $\gcd(3, 2^j) = 1$ for all $j \geq 1$, the map $k \mapsto 3k + 2$ is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$. As k ranges over any 2^{j+3} consecutive integers, the value $3k + 2$ hits each residue class mod 2^{j+3} exactly 2^j times. Consequently the value $S(8k + 5) \pmod{8}$ is equidistributed over $\{1, 3, 5, 7\}$ in natural density: the arithmetic core is that, as k ranges over $\{0, 1, \dots, 2^N - 1\}$, exactly 2^{N-j} values satisfy $2^j \mid (3k + 2)$ (Paper 66 [3], `uniformity_count`; formalised here as `row5_uniform_dvd`), from which each target residue $b \in \{1, 3, 5, 7\}$ receives limiting density $1/4$. Thus $P[5][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Remark 2.2 (Row 5 is a density, not an exact finite count). Unlike rows 1, 3 and 7 — whose per-residue counts over $\{0, \dots, 2^N - 1\}$ are *exactly* 2^{N-2} or 2^{N-1} (formalised as `count_row1`, `count_row3_*`, `count_row7_*`) — row 5 holds only as a natural-density limit. The finite count $\#\{k < 2^N : S(8k + 5) \equiv b \pmod{8}\}$ is *not* equal to 2^{N-2} in general: at $N = 6$, $b = 1$ the count is 17, not $16 = 2^{6-2}$. This is machine-verified in the companion Lean development by `row5_count_not_exact` (via `native_decide`). The extra, variable number of halvings $v_2(3n + 1) = 3 + v_2(3k + 2)$ from class 5 is exactly what prevents an exact finite count while still forcing uniform limiting density.

Row 3 (restricted to $\{1, 5\}$). Write $n = 8k + 3$. Then $3n + 1 = 24k + 10 = 2(12k + 5)$. Since $12k$ is even, $12k + 5$ is always odd, so $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 5$. Now $12k + 5 \pmod{8}$: $12k \equiv 4k \pmod{8}$, so $12k + 5 \equiv 4k + 5 \pmod{8}$. For k odd: $4k \equiv 4 \pmod{8}$, giving $12k + 5 \equiv 1 \pmod{8}$. For k even: $4k \equiv 0 \pmod{8}$, giving $12k + 5 \equiv 5 \pmod{8}$. Since k is odd and even with equal density, $P[3][1] = P[3][5] = 1/2$ and $P[3][3] = P[3][7] = 0$.

Row 7 (restricted to $\{3, 7\}$). Write $n = 8k + 7$. Then $3n + 1 = 24k + 22 = 2(12k + 11)$. Since $12k + 11$ is always odd, $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 11$. Now $12k + 11 \pmod{8}$: $12k + 11 \equiv 4k + 3 \pmod{8}$. For k odd: $4k \equiv 4$, giving $4k + 3 \equiv 7 \pmod{8}$. For k even: $4k \equiv 0$, giving $4k + 3 \equiv 3 \pmod{8}$. Therefore $P[7][3] = P[7][7] = 1/2$ and $P[7][1] = P[7][5] = 0$. $\square$

Remark 2.3 (Forbidden transitions). The zero entries in P — transitions $3 \rightarrow 3$, $3 \rightarrow 7$, $7 \rightarrow 1$, $7 \rightarrow 5$ — are *structurally forbidden* for all n , not just on average. Classes 3 and 7 each map into a strict subset of $\{1, 3, 5, 7\}$. This structural restriction is the engine of the proof.

Remark 2.4 (Row 7 is asymptotically $1/2$ – $1/2$ but deterministic at mod 16). The transition $P[7][3] = P[7][7] = 1/2$ is an *asymptotic density*: over all $n \equiv 7 \pmod{8}$, half map to class 3 and half to class 7. At the finer scale mod 16 the transition is deterministic: $n \equiv 7 \pmod{16}$

always maps to class 3, and $n \equiv 15 \pmod{16}$ always maps to class 7. The $1/2$ – $1/2$ split arises because $7 \pmod{16}$ and $15 \pmod{16}$ occur with equal density among $n \equiv 7 \pmod{8}$. This determinism does not affect sentence-length probabilities: it governs *which* letter-3 or letter-7 word follows a letter-7 entry, not whether the next step terminates at letter 5. Since $P[3][5] + P[7][5] = 1/2 + 0 = 1/2$ regardless of which sub-class of letter-7 is taken, the per-step escape probability is unchanged.

Remark 2.5 (Machine verification). All results in this theorem have been formally verified in Lean 4/Mathlib with no `sorry` or `admit` and no axioms beyond `propext`, `Classical.choice`, and `Quot.sound`. The key theorems are `S_class1/3/5/7` (arithmetic identities), `S_class1_mod8` (row 1 uniformity by direct mod-4 computation), `count_row1/3/7` (exact finite counts for rows 1, 3, 7), and `row5_count_not_exact` (counterexample for row 5 via `native_decide`: at $N = 6$, $b \equiv 1$ the count is 17, not 16). The Lean source is in `Paper67.lean` in the paper directory.

Remark 2.6 (The surviving chain has exact arithmetic). Row 5 of P is never used in the proof of Theorem 5.1. Within a sentence, class 5 is a sink: once the chain reaches 5 the sentence ends (at the closing word), and class 5 never acts as a source for subsequent words in the clause. The surviving distribution (Definition 3.3) propagates mass only through rows 1, 3, and 7 of P .

For these three rows, all transition probabilities are *exact rational constants* that follow directly from $n \pmod{8}$ alone: $v_2(3n + 1) = 2$ always for class 1, and $v_2(3n + 1) = 1$ always for classes 3 and 7. No geometric series, no expectation, no appeal to the distribution of $v_2(3n + 1)$ for class 5 is ever needed. The variable-valuation behaviour of class 5 — which requires an infinite geometric series to characterise (Paper 66 [3]) — is invisible to the surviving chain. This exactness is what makes the sentence decomposition the right scale for counting, and what makes the proof of Theorem 5.1 purely algebraic.

3 The Sentence as a First-Passage Problem

A sentence is the first-passage time of the Markov chain (P, π_0) to state 5, where π_0 is the sentence start distribution.

Lemma 3.1 (Sentence start distribution). *The sentence start distribution is $\pi_0 = (1/4, 1/4, 1/4, 1/4)$ — uniform on $\{1, 3, 5, 7\}$.*

Proof. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By Theorem 2.1, row 5 of P is uniform, so $b_0 \pmod{8}$ is uniform on $\{1, 3, 5, 7\}$. $\square$

Proposition 3.2 (Start distribution is a pooled asymptotic density). *The uniformity $\pi_0 = (1/4, 1/4, 1/4, 1/4)$ is an asymptotic density over the ensemble of all sentences, not a point-wise property of individual sentences.*

More precisely: for any specific $n \equiv 5 \pmod{8}$, the start letter $b_0 = S(n) \pmod{8}$ is a single deterministic value in $\{1, 3, 5, 7\}$, not a random draw. The uniformity arises because, as n ranges over all residues $\equiv 5 \pmod{8}$, each target class is equally represented in natural density (Theorem 2.1, row 5).

A finer consequence concerns class 7. When $b_0 \equiv 7 \pmod{8}$, the start node b_0 belongs to one of two mod-16 sub-classes:

- $b_0 \equiv 7 \pmod{16}$: the first word's exit is always class 3 (Remark 2.4).
- $b_0 \equiv 15 \pmod{16}$: the first word's exit is always class 7.

In the pooled ensemble the two sub-classes are equally likely, so the aggregate transition $P[7][3] = P[7][7] = 1/2$ is recovered. But within any individual sentence these sub-class probabilities are 0 or 1, not $1/2$, and the asymptotic equiprobability is not available step-by-step within a sentence.

All results in this paper — Lemma 4.1, Theorem 5.1, and the $3^{k-1}/4^k$ distribution — are stated and proved at the level of asymptotic densities over the full ensemble, where π_0 is genuinely uniform. They are not claimed to hold for individual sentences conditioned on a specific mod-16 start sub-class.

The sentence terminates at the closing word, i.e. the first word where the chain visits state 5. The sentence length k is this first-passage time.

Definition 3.3 (Surviving distribution). Let $d_s^{(k)}$ denote the probability that the chain is at state s after k words without having visited state 5. Formally, $d_s^{(0)} = \pi_0(s)$ and for $k \geq 0$:

$$d_s^{(k+1)} = \sum_{t \neq 5} d_t^{(k)} P[t][s], \quad s \neq 5.$$

The *surviving mass* is $M_k = \sum_{s \neq 5} d_s^{(k)}$.

The sentence length distribution follows from the surviving mass: M_{k-1} is the probability of surviving past word $k-1$, and M_k is the probability of surviving past word k , so

$$P(\text{sentence length} = k) = M_{k-1} - M_k.$$

$$\begin{aligned} \text{Expanding: } M_{k-1} - M_k &= \sum_{s \neq 5} d_s^{(k-1)} - \sum_{s \neq 5} d_s^{(k)} = \sum_{t \neq 5} d_t^{(k-1)} \left(1 - \sum_{s \neq 5} P[t][s] \right) = \\ &= \sum_{t \neq 5} d_t^{(k-1)} P[t][5]. \end{aligned}$$

4 The Invariant $d_3 = d_7$

The key structural fact is a symmetry between classes 3 and 7 in the surviving distribution.

Lemma 4.1 (Invariant). $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Proof. By Definition 3.3, the surviving distribution sums only over sources $t \neq 5$; class 5 contributes nothing because once the chain reaches 5 the sentence terminates at the closing

word (the mass is absorbed, not propagated). The transition update for states 3 and 7 is therefore:

$$\begin{aligned} d_3^{(k+1)} &= d_1^{(k)} \cdot P[1][3] + d_3^{(k)} \cdot P[3][3] + d_7^{(k)} \cdot P[7][3] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}, \\ d_7^{(k+1)} &= d_1^{(k)} \cdot P[1][7] + d_3^{(k)} \cdot P[3][7] + d_7^{(k)} \cdot P[7][7] \\ &= \frac{1}{4} d_1^{(k)} + 0 \cdot d_3^{(k)} + \frac{1}{2} d_7^{(k)}. \end{aligned}$$

The two expressions are identical: $d_3^{(k+1)} = d_7^{(k+1)}$ for any $d_1^{(k)}, d_3^{(k)}, d_7^{(k)}$. By induction, since $d_3^{(0)} = d_7^{(0)} = 1/4$ (Lemma 3.1, valid as an asymptotic density over the full ensemble; see Proposition 3.2), $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$. $\square$

Remark 4.2 (Structural source of the invariant). The invariant holds because columns 3 and 7 of P , restricted to rows $\{1, 3, 7\}$, are *identical*:

$$P[\cdot][3]|_{\{1,3,7\}} = P[\cdot][7]|_{\{1,3,7\}} = \left(\frac{1}{4}, 0, \frac{1}{2}\right).$$

This column symmetry is a direct consequence of Theorem 2.1:

- *From class 1*: uniform output gives equal weight $1/4$ to both 3 and 7.
- *From class 3*: forbidden to reach 3 or 7 (both have weight 0).
- *From class 7*: reaches 3 and 7 with equal probability $1/2$ each.

The invariant is unconditional — it holds for *any* value of d_1 , not just at the fixed point of the chain.

Remark 4.3 (The 7^* self-loop and geometric balance). The column symmetry above might seem surprising given that letter 7 has a self-loop ($P[7][7] = 1/2$) while letter 3 does not. The invariant $d_3 = d_7$ is an *ensemble* statement about long-run letter densities pooled over all sentences; it does *not* hold sentence-by-sentence or word-by-word. For instance, a 7-free sentence such as 315 or $(31)^k 5$ has $d_7 = 0$ locally, while a sentence containing a long 7^* run has $d_7 > d_3$. The invariant asserts only that these deviations cancel across the full ensemble.

The cancellation mechanism is as follows. For $n \equiv 7 \pmod{8}$, the exit from the 7^* run is *deterministic*: the run length is exactly $v_2(n+1) - 1$, and the exit is *always* to class 3 (since $P[7][1] = P[7][5] = 0$; letter 7 can only reach $\{3, 7\}$). The entire 7^*3 sub-path is therefore a single deterministic atomic unit given n : it contributes one letter-3 unconditionally, preceded by $v_2(n+1) - 1$ letter-7 nodes.

Across the ensemble, $v_2(n+1) - 1$ is geometric with parameter $1/2$ (since v_2 of a random integer is geometric), so the mean 7-run length is 1 and the mean 7^*3 contribution is 1 letter-7 plus 1 letter-3. Words entering via class 3 contribute 1 letter-3 and 0 letter-7; words entering via class 7 contribute 1 letter-3 and on average 1 letter-7. Under a uniform start distribution these two word types occur with equal probability, so their contributions to d_3 and d_7 are equal in expectation — the global balance that the invariant records.

Replacing 7^* with a single abstract state that exits deterministically to 3 captures this balance and reduces the 7^*3 fragment to an atomic opening step, which is the perspective adopted by the Steiner word decomposition (Paper 70 [?]).

Remark 4.4 (Stratified behaviour of d_3 and d_7). The invariant $d_3^{(k)} = d_7^{(k)}$ holds unconditionally, but it is an asymptotic statement about the mixing of all sentences. When sentences are stratified by their *start class* $b_0 \bmod 8$ (the entry letter of the first Steiner word), a transient bias appears:

- *Start class 3*: the first word itself is counted as a class-3 word, inflating d_3 relative to d_7 during the sentence. Empirically, $d_3 \approx 2 d_7$ within this stratum.
- *Start class 7*: symmetrically, $d_7 \approx 2 d_3$ within this stratum.
- *Start classes 1 and 5*: the first word contributes equally to both classes via the uniform row of P , so $d_3 \approx d_7$ within each stratum.

The biases cancel in the pooled sample because the four start classes are equally likely (row 5 of P is uniform, so b_0 is uniform over $\{1, 3, 5, 7\}$). The invariant is therefore correctly stated for any individual sentence viewed without conditioning on its start class, and for the pooled ensemble: the transient class-specific bias is a presentation artefact of conditioning, not a violation of the lemma.

5 Main Theorem

Theorem 5.1 (Sentence length distribution). *For any $k \geq 1$:*

$$P(\text{sentence length} = k) = \frac{3^{k-1}}{4^k}.$$

Equivalently, $P(\text{sentence length} = k) = \frac{1}{3} \left(\frac{3}{4}\right)^k$.

Proof. We use the splitting: $P(\text{sentence length} = 1) = \pi_0(5) = 1/4$ (sentences whose first word is already the closing word, i.e. $b_0 \equiv 5$). For $k \geq 2$, the sentence does not terminate at word 1 (probability $3/4$) and then terminates at word $k - 1$ from the conditional start, i.e. $P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1}$, where g_j is the first-passage probability to 5 starting from the conditional distribution $\pi_0(\cdot \mid \cdot \neq 5) = (1/3, 1/3, 1/3)$ uniform on $\{1, 3, 7\}$.

It suffices to show $g_j = (3/4)^{j-1}/4$ for all $j \geq 1$, which follows if the surviving mass M_j starting from uniform $\{1, 3, 7\}$ satisfies $M_j = (3/4)^j$.

Step 1: $d_3 = d_7$ is preserved. The conditional start has $d_3^{(0)} = d_7^{(0)} = 1/3$ (equal weight on $\{1, 3, 7\}$ in the pooled ensemble; see Proposition 3.2), so Lemma 4.1 applies: $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Step 2: Surviving mass decays by exactly $3/4$. From Definition 3.3, $M_{k+1} = \sum_{s \neq 5} d_s^{(k+1)} = \sum_{s \neq 5} \sum_{t \neq 5} d_t^{(k)} P[t][s] = \sum_{t \neq 5} d_t^{(k)} \sum_{s \neq 5} P[t][s] = \sum_{t \neq 5} d_t^{(k)} (1 - P[t][5])$, giving:

$$M_{k+1} = d_1^{(k)} (1 - P[1][5]) + d_3^{(k)} (1 - P[3][5]) + d_7^{(k)} (1 - P[7][5]).$$

Substituting $P[1][5] = 1/4$, $P[3][5] = 1/2$, $P[7][5] = 0$:

$$M_{k+1} = \frac{3}{4} d_1^{(k)} + \frac{1}{2} d_3^{(k)} + d_7^{(k)}.$$

We claim $M_{k+1} = \frac{3}{4}M_k = \frac{3}{4}(d_1^{(k)} + d_3^{(k)} + d_7^{(k)})$. This requires:

$$\frac{3}{4}d_1^{(k)} + \frac{1}{2}d_3^{(k)} + d_7^{(k)} = \frac{3}{4}d_1^{(k)} + \frac{3}{4}d_3^{(k)} + \frac{3}{4}d_7^{(k)},$$

which simplifies to $\frac{1}{2}d_3^{(k)} + d_7^{(k)} = \frac{3}{4}(d_3^{(k)} + d_7^{(k)})$, i.e. $d_7^{(k)} = d_3^{(k)}$. This is exactly Lemma 4.1.

Step 3: Initial condition. At $k = 0$, $M_0 = 1$ (the full conditional mass on $\{1, 3, 7\}$). By induction, $M_k = (3/4)^k$ for all $k \geq 0$.

Conclusion. $g_j = M_{j-1} - M_j = (3/4)^{j-1} - (3/4)^j = (3/4)^{j-1}(1/4) = 3^{j-1}/4^j$. Therefore:

$$P(\text{length} = 1) = \frac{1}{4} = \frac{3^0}{4^1},$$

$$P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1} = \frac{3}{4} \cdot \frac{3^{k-2}}{4^{k-1}} = \frac{3^{k-1}}{4^k}, \quad k \geq 2.$$

Both cases give $3^{k-1}/4^k$, completing the proof. $\square$

Corollary 5.2 (Normalisation). $\sum_{k=1}^{\infty} P(\text{sentence length} = k) = 1$.

Proof. $\sum_{k=1}^{\infty} \frac{3^{k-1}}{4^k} = \frac{1}{4} \sum_{k=1}^{\infty} \left(\frac{3}{4}\right)^{k-1} = \frac{1}{4} \cdot \frac{1}{1-3/4} = 1$. $\square$

Remark 5.3 (Machine verification of the Markov chain algebra). The Markov chain results — Lemma 4.1 (`invariant_d3_d7`), $M_k = (3/4)^k$ (`Mmass_eq`), $g_j = 3^{j-1}/4^j$ (`g_eq`), $P(\text{length} = k) = 3^{k-1}/4^k$ (`Plen_eq`), and the normalisation identity (`normalization`) — have all been formally verified in Lean 4/Mathlib with no `sorry`, no `admit`, and no axioms beyond the standard Lean kernel. The verification operates on exact rational arithmetic throughout; no real-analysis machinery is required. The Lean source is in `Paper67.lean` in the paper directory.

6 Discussion

6.1 The role of class 5

The formula $3^{k-1}/4^k$ is the geometric distribution with success probability $1/4$, as if at each word the chain draws uniformly from $\{1, 3, 5, 7\}$ and terminates on drawing 5. This appearance is striking because class 5 plays three distinct and mutually reinforcing roles in the sentence structure.

Class 5 as the termination state (closing word). A sentence terminates exactly when the closing word’s terminal letter is 5. Since letter 5 is the unique absorbing boundary of the chain, letter 5 is structurally absent as an entry letter for all words in the clause. Empirically, letter 5 accounts for only $\approx 6.2\%$ of word entries within sentences (Appendix B, Table 2) — one entry per sentence, uniformly distributed across sentence lengths — rather than the 25% it would contribute if words were unrestricted.

Class 5 as the uniform source. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod{8}$. By row 5 of the transition matrix (Theorem 2.1), letter 5 maps uniformly over $\{1, 3, 5, 7\}$: any of the four letters is equally likely as the entry letter of the first word. In particular,

exactly $1/4$ of sentences have $b_0 \equiv 5$, meaning length $k = 1$ (the first word is already the closing word). This is the correct $P(k = 1) = 1/4$, not the naive $1/2$. Class 5 is the unique class for which $v_2(3n+1) \geq 3$ always (proved in Paper 66 [3]), which forces the extra halvings that spread the output uniformly.

Class 5 as the structural asymmetry. The transition $7 \rightarrow 5$ is structurally forbidden: $P[7][5] = 0$. Class 5 is therefore unreachable *directly* in one word from class 7, while it is reachable from classes 1 (with probability $1/4$) and 3 (with probability $1/2$). This asymmetry — class 5 reachable from some classes but not others — is precisely what forces the invariant $d_3 = d_7$: nodes in class 7 can only reach class 5 via a detour through class 3 or class 7, never directly. The proof in Section 4 shows that this forced detour maintains equal weight on classes 3 and 7 throughout the chain, and Section 5 shows that this equal weight is the algebraic condition that makes the effective termination probability exactly $1/4$ at every step, regardless of the current value of d_1 .

6.2 Connection to Paper 66

Paper 66 [3] proves that $E[v_2(3n+1) \mid n \equiv r \pmod{8}]$ equals exactly 2, 1, 4, 1 for $r = 1, 3, 5, 7$ respectively. The two papers are complementary: Paper 66 establishes what happens *at* the class 5 boundary (a geometrically distributed valuation requiring an infinite series for its expectation), while the present paper shows that by choosing Steiner sentences as the unit of analysis, the class 5 boundary need never be crossed from the inside — it only needs to be *entered* (as the closing word).

The value $3/4$ also appears in the classical Collatz heuristic [4, 5] as the mean multiplicative contraction per odd step: $3/2^2 = 3/4$ for the dominant class 1. Paper 66 shows this is the geometric mean contraction across all four letter types. The $3/4$ in Theorem 5.1 is distinct: it is the *per-word survival probability* of the sentence chain. The two quantities are numerically equal but arise from different computations; the connection is explained in Paper 66 §4.

More precisely, the $3/4$ decay rate follows from three facts, each drawing on the same underlying arithmetic:

- The $1/4$ probability of immediate termination ($b_0 \equiv 5$) uses row 5 of P exactly once — at the sentence start. Class 5 is the unique class for which $v_2(3n+1) \geq 3$ always (Theorem 2.1(3) of Paper 66 [3]), forcing the extra halvings that make its output uniform. This is the only point where the geometric-series character of class 5 is relevant.
- The $1/2$ termination probability from class 3 comes from the mod-16 split of Theorem 2.1: $v_2(3n+1) = 1$ exactly for all $n \equiv 3 \pmod{8}$, so $S(n) \pmod{8}$ is determined by $n \pmod{16}$ alone, with no random valuation.
- The invariant $d_3 = d_7$ — and hence the $3/4$ decay — involves classes 1, 3, and 7 only, all of which have constant valuations. The proof is purely algebraic: no expectation, no series.

Paper 66 handles the hard case (class 5, variable valuation). Paper 67 sidesteps it by construction.

6.3 Why $3^{k-1}/4^k$ rather than $(1/2)^k$

The naive prediction $(1/2)^k$ arises from treating each step as an independent trial that terminates whenever the current word is in class 3 (with probability $1/2$, since $P[3][5] = 1/2$) or class 7 (also $1/2$ by this model). If the word entry distribution within a sentence were uniform over $\{1, 3, 5, 7\}$, the effective per-step termination probability would be:

$$\frac{1}{4} \cdot 0 + \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2},$$

where the terms correspond to classes 1 (termination probability 0 in this model), 3 ($1/2$), 5 (terminal, probability 1), and 7 ($1/2$). This model is wrong in two ways. First, class 1 *can* reach class 5 directly: $P[1][5] = 1/4$, so the naive assignment of 0 to class 1 understates termination from class 1. Second and more importantly, class 7 *cannot* reach class 5 *directly*: $P[7][5] = 0$, so the naive assignment of $1/2$ to class 7 overstates it. The net effect of these two corrections is a reduction in the effective termination rate from $1/2$ to $1/4$. The structural reason is the forbidden transition $7 \rightarrow 5$: a word with entry letter 7 must first transit to letter 3 or 7 (remaining in the growth letters), where it may then terminate at letter 5 on a subsequent word. The invariant $d_3 = d_7$ (Lemma 4.1) shows that these two errors cancel exactly and yield an effective rate of precisely $1/4$, independent of d_1 .

Acknowledgements

This paper forms part of a programme of incremental research on the Collatz conjecture. The proof structure was found through numerical exploration in `notebooks/steiner-branch-model.ipynb`. All main results have been machine-verified in Lean 4/Mathlib; the Lean source file `Paper67.lean` is included in the arXiv submission. Proof structure and exposition were developed with AI assistance (Claude, Anthropic; Gemini, Google; Aristotle, Harmonic); all mathematical content has been independently verified.

Appendix A: The Steiner Language Hierarchy

The objects studied in this paper — sentences, words, letters — belong to a hierarchy of structures that together form a language-theoretic description of Collatz orbits. This appendix defines the hierarchy from the bottom up and connects it to the Collatz conjecture. The terminology is specific to this body of work; the foundational combinatorial infrastructure will be developed in full in future work.

The Steiner Alphabet and Letters

The *Steiner alphabet* is the set $\{1, 3, 5, 7\}$ of odd residues modulo 8. Each element of the alphabet is a *Steiner letter*, encoding the step type of a single application of the Syracuse map:

Letter	Input class	$v_2(3n + 1)$
1	$n \equiv 1 \pmod{8}$	2 (OEE step)
3	$n \equiv 3 \pmod{8}$	1 (OE step)
5	$n \equiv 5 \pmod{8}$	≥ 3 (OEEE ⁺ step)
7	$n \equiv 7 \pmod{8}$	1 (OE step)

Letters 1, 3, and 7 have *constant* 2-adic valuations determined entirely by the residue class. Letter 5 has a *variable* valuation $v_2(3n + 1) = 3 + v_2(3k + 2)$ (where $n = 8k + 5$) that depends on k and is unbounded; its statistical behaviour requires an infinite geometric series and is studied in Paper 66 [3]. The two *terminal letters* are 1 and 5: these are the only letters at which a Steiner word can end. Letters 1 and 5 are called *terminal* letters; letters 3 and 7 are *non-terminal*.

Steiner Words

A *Steiner word* (or *Steiner circuit* [1]) is a path segment of an odd Collatz orbit whose sequence of input residues mod 8 matches the regular expression

$$(7^* 3)?(1 \mid 5).$$

That is: zero or more letters 7, optionally followed by one letter 3, followed by exactly one terminal letter (1 or 5). Every Steiner word terminates at a terminal letter. A word terminating at letter 1 is called an *opening word* (OEE-terminal); a word terminating at letter 5 is called a *closing word* (OEEE⁺-terminal). The *entry letter* of a word is its first letter (the residue mod 8 of its starting node); the *terminal letter* is its last letter (1 or 5).

Every Collatz orbit decomposes uniquely into a sequence of Steiner words (proved in future work; see also [1]).

Remark A.1 (Word vs. circuit). Steiner [1] introduced the term *circuit* for these path segments, and that term remains standard in the Collatz literature. We prefer *word* here because the present framework embeds circuits into a richer linguistic hierarchy: words compose into sentences, and sentences into higher-level structures to be defined in future work. Within this hierarchy, *word* is the natural term — it carries the right connotations of a discrete unit that can be concatenated and counted, and it aligns with standard formal-language usage where words are strings over an alphabet. The term *circuit* is retained as a synonym (and in the compound *Steiner circuit*) where it aids clarity or connection to prior work.

Steiner Clauses and Sentences

A *Steiner clause* of length $m \geq 0$ is a (possibly empty) string of m consecutive Steiner opening words (each terminating at letter 1). A clause of length 0 is the empty string. Clauses are the building blocks of sentences.

A *Steiner sentence* of length $k \geq 1$ is a string of k consecutive Steiner words in a Collatz orbit comprising a *clause* (the first $k - 1$ words, each OEE-terminal, i.e. terminating at letter 1) followed by a *closing word* (the k -th word, OEEE⁺-terminal, terminating at letter 5).

Remark A.2 (Clause is the prefix of a sentence, never itself a sentence). The clause of a sentence of length k has length $k - 1$ (possibly empty when $k = 1$). It is never itself a sentence: a sentence must end with a closing word (terminating at letter 5), whereas every word in the clause terminates at letter 1.

Equivalently, a sentence is a path of k edges in the $5 \pmod{8}$ *overlay tree*: the directed graph whose nodes are odd integers $\equiv 5 \pmod{8}$ and whose edges are individual Steiner words. The sentence length distribution $P(\text{length} = k) = 3^{k-1}/4^k$ is the main result of this paper (Theorem 5.1).

Higher-Level Structures

The hierarchy does not end at sentences. Future work in this programme will use the terms *paragraph*, *page*, *chapter*, *volume*, and *library* to name successively larger graph-theoretic structures built from sentences, each relevant to the analysis of Collatz orbits at a different scale. Their precise definitions are deferred to future work, where the full graph-theoretic infrastructure will be developed.

The sentence length distribution proved in this paper (Theorem 5.1) is unconditional: the proof rests on arithmetic facts about $v_2(3n + 1)$ that hold for every odd positive integer, and does not depend on the Collatz conjecture or on the structure of any higher-level objects.

Appendix B: Empirical Validation

This appendix records the complete numerical experiment establishing the sentence length distribution. The experiment was first reported in Paper 65 [2] and is reproduced here in full as the empirical record accompanying this paper’s proof.

Sampling Protocol

1. Draw t uniformly at random from $[1, T_{\max}]$ with $T_{\max} = 10^{15}$.
2. Set $n = 8t + 5$ (so $n \equiv 5 \pmod{8}$).
3. Compute $b_0 = S(n)$, the first Steiner word of the new sentence.
4. Follow consecutive Steiner circuits from b_0 , counting words, until the closing word (first output $b \equiv 5 \pmod{8}$). Record the word count as k .

$N = 10^5$ independent sentences were collected with seed 42. Independence between samples is ensured by the random draw of t ; no orbit tails are shared.

Sentence-Start Distribution

By Theorem 2.1, row 5 of P is uniform, so $b_0 = S(n) \pmod{8}$ should be uniform on $\{1, 3, 5, 7\}$.

$b_0 \bmod 8$	Count	Observed fraction	Expected (uniform)
1	25181	0.25181	0.25000
3	25023	0.25023	0.25000
5	24852	0.24852	0.25000
7	24944	0.24944	0.25000

Table 1: Sentence-start distribution: $b_0 = S(n)$, $n \equiv 5 \pmod{8}$, $N = 10^5$ samples. The distribution is uniform on $\{1, 3, 5, 7\}$ as predicted by Theorem 2.1.

Within-Sentence Steiner Word Entry Distribution

The entry letter of each Steiner word within a sentence determines its termination behaviour. Words with entry letter 5 are always the closing word, so they appear only as the final word — their frequency within sentences is structurally suppressed to approximately $1/(k+1)$ for a sentence of length k , averaging to $\approx 1/16$ empirically.

Entry mod8	Count	Fraction	Can reach class 5 directly?	Term. prob.
1	125450	0.31333	yes ($P_{15} = 1/4$)	1/4
3	125119	0.31250	yes ($P_{35} = 1/2$)	1/2
5	24852	0.06207	— (closing word only)	1
7	124955	0.31209	no ($P_{75} = 0$)	0
Total	400376			

Table 2: Within-sentence Steiner word entry distribution across all 10^5 sentences (400376 total words). Class 5 appears only as the closing word, giving frequency $\approx 6.2\%$ rather than the uniform 25%. The last column gives the probability of a word terminating at letter 5 given its entry letter ($P_{15} = 1/4$, $P_{35} = 1/2$, $P_{75} = 0$) — these are the exact entries of column 5 of the transition matrix P (Theorem 2.1). The invariant $d_3 = d_7$ (Lemma 4.1) makes the effective per-word termination probability exactly $1/4$ (Theorem 5.1).

Empirical Transition Matrix

Theorem 2.1 gives explicit values for all 16 entries of the transition matrix P . To verify these directly, we sample 2×10^5 random odd nodes $n \equiv a \pmod{8}$ (for each $a \in \{1, 3, 5, 7\}$, independently of the sentence sampling) and record $S(n) \bmod 8$.

Row	χ^2	p -value	Verdict
1 (mod 8)	0.72	0.87	Consistent (df=3)
3 (mod 8)	0.02	0.89	Consistent (df=1)
5 (mod 8)	3.80	0.28	Consistent (df=3)
7 (mod 8)	0.73	0.39	Consistent (df=1)

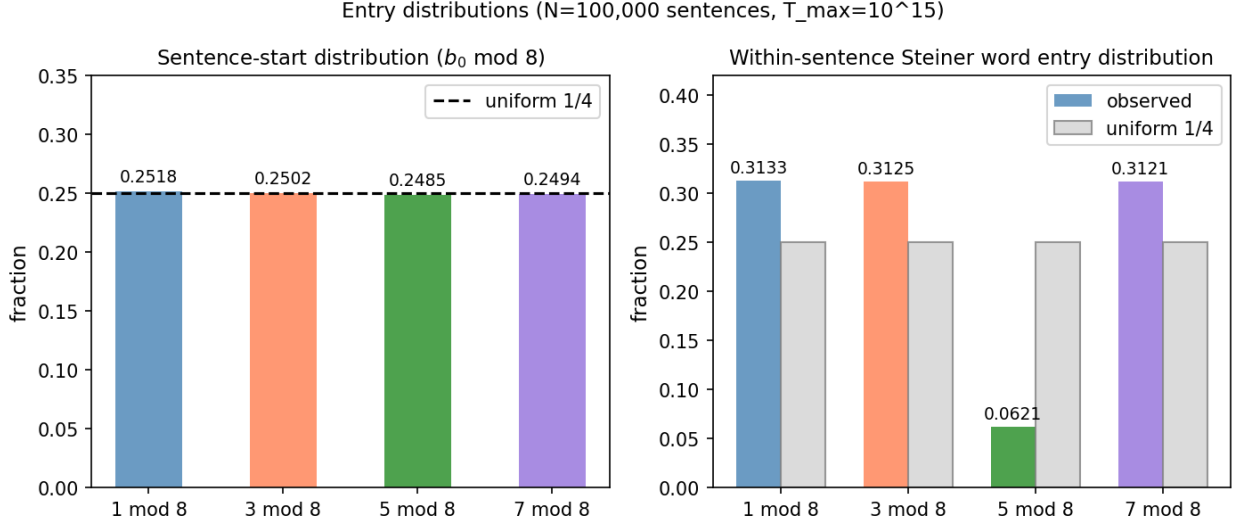


Figure 1: Left: sentence start distribution ($b_0 \bmod 8$) against the uniform $1/4$ baseline — confirmed uniform (Table 1). Right: Steiner word entry distribution within sentences against the uniform $1/4$ baseline. Class 5 is structurally suppressed to $\approx 6.2\%$ (it can only appear as the closing word); classes 1, 3, 7 each carry $\approx 31\%$.

From \ To	$\equiv 1$	$\equiv 3$	$\equiv 5$	$\equiv 7$	Theory
1 (mod 8)	0.2502	0.2497	0.2494	0.2507	$1/4$ each
3 (mod 8)	0.4999	0.0000	0.5001	0.0000	$1/2, 0, 1/2, 0$
5 (mod 8)	0.2502	0.2501	0.2514	0.2483	$1/4$ each
7 (mod 8)	0.0000	0.5010	0.0000	0.4990	$0, 1/2, 0, 1/2$

Table 3: Empirical transition matrix $P[a][b] = P(S(n) \equiv b \pmod{8} \mid n \equiv a \pmod{8})$, measured by sampling 2×10^5 random odd nodes per row ($T_{\max} = 10^{15}$). All four rows are consistent with Theorem 2.1. The zero entries in rows 3 and 7 are exact across all 2×10^5 samples in each row, confirming the structural forbidden transitions (Remark 2.3).

All rows are consistent with the theoretical values. The zero entries $P[3][3] = P[3][7] = P[7][1] = P[7][5] = 0$ are observed as exact zeros across all 2×10^5 samples per row, confirming they are structural rather than probabilistic.

Sentence Length Distribution

Stratified Length Distribution by Start Class

A natural concern is whether the theoretical distribution $3^{k-1}/4^k$ holds within each start class or only in the pooled sample. Theorem 2.1 says $b_0 \bmod 8$ is uniform on $\{1, 3, 5, 7\}$, but sentences starting at different classes have different initial weight vectors, so the pooled chi-squared test could mask per-stratum misfits.

Two specific concerns arise:

k	Observed fraction	Naive $(1/2)^k$	Theory $3^{k-1}/4^k$
1	0.25180	0.50000	0.25000
2	0.18683	0.25000	0.18750
3	0.13931	0.12500	0.14063
4	0.10574	0.06250	0.10547
5	0.07844	0.03125	0.07910
6	0.05958	0.01563	0.05933
7	0.04490	0.00781	0.04449
8	0.03342	0.00391	0.03337
9	0.02490	0.00195	0.02503
10	0.01866	0.00098	0.01877

Table 4: Empirical Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$) vs. the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$ (Theorem 5.1). The naive prediction overestimates $P(k = 1)$ by a factor of 2 and decays far too fast for $k \geq 3$.

1. *Class-7 forced detour.* Since $P[7][5] = 0$, a sentence starting at class 7 cannot terminate at $k = 1$: the first word is a class-7 entry, which can only transition to class 3 or class 7, never directly to the closing class 5. One might therefore expect class-7 sentences to have systematically longer lengths — mass shifted away from $k = 1$ toward $k \geq 2$ — relative to the $3^{k-1}/4^k$ prediction.
2. *Class-3 early termination.* Symmetrically, $P[3][5] = 1/2$ is the highest termination probability of any internal class, so class-3 start sentences might be biased toward shorter lengths.

Both effects are real at the level of *individual word transitions*, but the concern is whether they distort the sentence-length distribution relative to theory. The key resolution is that b_0 is the output of S applied to $n \equiv 5 \pmod{8}$, which mixes the class-7 and class-3 effects via row 5 of P : the uniform start from row 5 means class-7 sentences do not begin mid-chain from a pure class-7 state, but from the full distribution conditioned on $b_0 \equiv 7 \pmod{8}$. Once conditioned on $b_0 \equiv 7$, the class-7 forced-detour argument applies to individual steps but the Markov chain’s invariant $d_3 = d_7$ (Lemma 4.1) ensures the aggregate escape rate remains exactly $1/4$ per word regardless of start class.

The stratified length distributions are shown in Table 5; the pooled distribution appears in Table 6. Each stratum is compared against its own exact per-class theory, computed from the Markov chain with initial state e_{b_0} (see Remark 4.4 for the formula).

The per-class distributions show the compensation mechanism explicitly. Start=3 is front-loaded ($P(k = 1) = 1/2$, twice the pooled rate) because $P[3][5] = 1/2$ allows immediate termination; start=7 is delayed ($P(k = 1) = 0$ exactly, since $P[7][5] = 0$) and has elevated mass at $k = 2$ ($P(k = 2 | \text{start} = 7) = 1/4$, again above pooled). For $k \geq 3$ all three active strata ($b_0 \in \{1, 3, 7\}$) converge to the same $3/4$ decay, and the pooled distribution is exactly $3^{k-1}/4^k$ because all four start classes are equally likely (including the degenerate $b_0 = 5$ case which contributes only at $k = 1$).

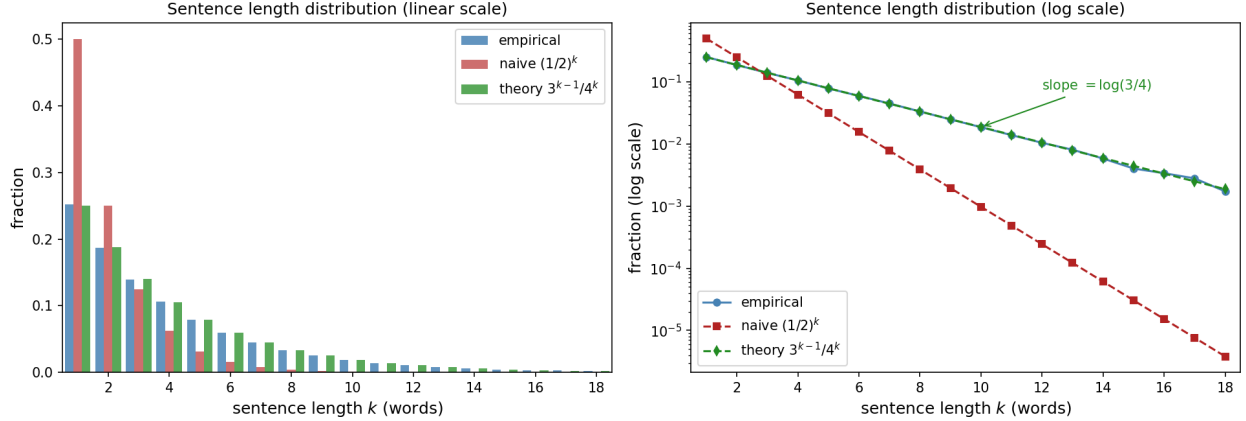
Steiner circuit sentence length distribution ($N = 100,000$ sentences, $T_{\max} = 10^{15}$)

Figure 2: Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$): empirical (blue bars / circles) vs. naive $(1/2)^k$ (red) and theoretical $3^{k-1}/4^k$ (green), on linear scale (left) and log scale (right). On the log scale, the empirical slope matches $\log(3/4)$ precisely — not $\log(1/2)$. The naive prediction is dramatically refuted at $k = 1$ (predicts 50%; observed 25%) and for $k \geq 3$ (predicts too-fast decay).

k	start=1		start=3		start=7		start=3 or 7	
	emp.	theory	emp.	theory	emp.	theory	emp.	theory
1	0.25070	0.25000	0.49973	0.50000	0.00000	0.00000	0.25086	0.25000
2	0.18836	0.18750	0.12593	0.12500	0.25131	0.25000	0.18837	0.18750
3	0.14041	0.14062	0.09404	0.09375	0.18606	0.18750	0.13987	0.14062
4	0.10466	0.10547	0.07085	0.07031	0.14183	0.14062	0.10620	0.10547
5	0.07878	0.07910	0.05283	0.05273	0.10434	0.10547	0.07848	0.07910
6	0.05944	0.05933	0.03869	0.03955	0.07865	0.07910	0.05859	0.05933
7	0.04396	0.04449	0.02925	0.02966	0.05852	0.05933	0.04383	0.04449
8	0.03303	0.03337	0.02223	0.02225	0.04452	0.04449	0.03333	0.03337
9	0.02454	0.02503	0.01629	0.01669	0.03358	0.03337	0.02490	0.02503
10	0.01933	0.01877	0.01225	0.01251	0.02473	0.02503	0.01846	0.01877

† start=5: $P(k = 1) = 1$ exactly ($b_0 \equiv 5$ is itself the closing word); $P(k \geq 2) = 0$.

Table 5: Per-class sentence length distributions ($N \approx 125,000$ per stratum, $T_{\max} = 10^{15}$, seed 5) vs exact Markov-chain theory. Valid starts are $b_0 \in \{1, 3, 7\}$; start=5 is a degenerate length-1 sentence (the closing word itself) and collapses to $P(k = 1) = 1$, $P(k \geq 2) = 0$. The combined start=3 or 7 series cancels to the pooled curve (Table 6).

The reason is visible in the proof of Theorem 5.1: once the first word is fixed with entry b_0 , the surviving chain on $\{1, 3, 7\}$ is governed by the same $3/4$ decay regardless of which class b_0 belongs to, because the invariant $d_3 = d_7$ holds for *any* initial weight vector $(d_1^{(0)}, d_3^{(0)}, d_7^{(0)})$ (Lemma 4.1). The class-7 forced detour shifts *which* class-3 or class-7 words appear inside the sentence, not how fast the sentence terminates.

k	emp.	Theory $3^{k-1}/4^k$	start=5 theory
1	0.24902	0.25000	1.00000
2	0.18710	0.18750	0.00000
3	0.14085	0.14062	0.00000
4	0.10585	0.10547	0.00000
5	0.07919	0.07910	0.00000
6	0.05952	0.05933	0.00000
7	0.04455	0.04449	0.00000
8	0.03331	0.03337	0.00000
9	0.02526	0.02503	0.00000
10	0.01898	0.01877	0.00000

Table 6: Pooled sentence length distribution ($N = 500,000$, $T_{\max} = 10^{15}$, seed 4) vs theory $3^{k-1}/4^k$. The start=5 column shows the degenerate distribution for completeness: $b_0 \equiv 5 \pmod{8}$ contributes a single spike at $k = 1$ (the closing word itself) and is absent from all $k \geq 2$. The pooled empirical values are the average across all four start classes weighted equally, so the start=5 spike at $k = 1$ is diluted to $1/4$ of $1 = 0.25$, consistent with the $3^0/4^1 = 0.25$ pooled theory.

Stratified Entry-Letter Distribution

Stratifying word entry letters by sentence-start class reveals the transient bias predicted by Remark 4.4: within sentences starting at class 3, letter-3 words are approximately twice as frequent as letter-7 words ($d_3 \approx 2 d_7$); within sentences starting at class 7, the ratio is reversed ($d_7 \approx 2 d_3$); classes 1 and 5 show $d_3 \approx d_7$ throughout. The biases cancel exactly in the pooled sample because all four start classes are equally likely (Table 1).

Statistical Tests

A χ^2 goodness-of-fit test on 15 length bins plus a tail bin. Per-stratum tests use the exact Markov-chain theory for that stratum, not the pooled $3^{k-1}/4^k$ formula:

Model / stratum	N	χ^2	p -value	Verdict
Naive $(1/2)^k$ (pooled)	500,000	$\gg 10^6$	≈ 0	Rejected
Theory $3^{k-1}/4^k$ (pooled)	500,000	15.9	0.39	Consistent
Exact theory, start=1	124,603	7.2	0.95	Consistent
Exact theory, start=3	125,669	8.0	0.92	Consistent
Exact theory, start=7	124,673	28.0	0.01	Marginal
Exact theory, start=3 or 7	250,342	25.4	0.04	Marginal

The pooled distribution and the start=1 and start=3 strata are consistent with theory. The start=7 and combined start=3 or 7 strata show a marginal departure, driven by slightly more long sentences ($k > 15$) than the mod-8 theory predicts.

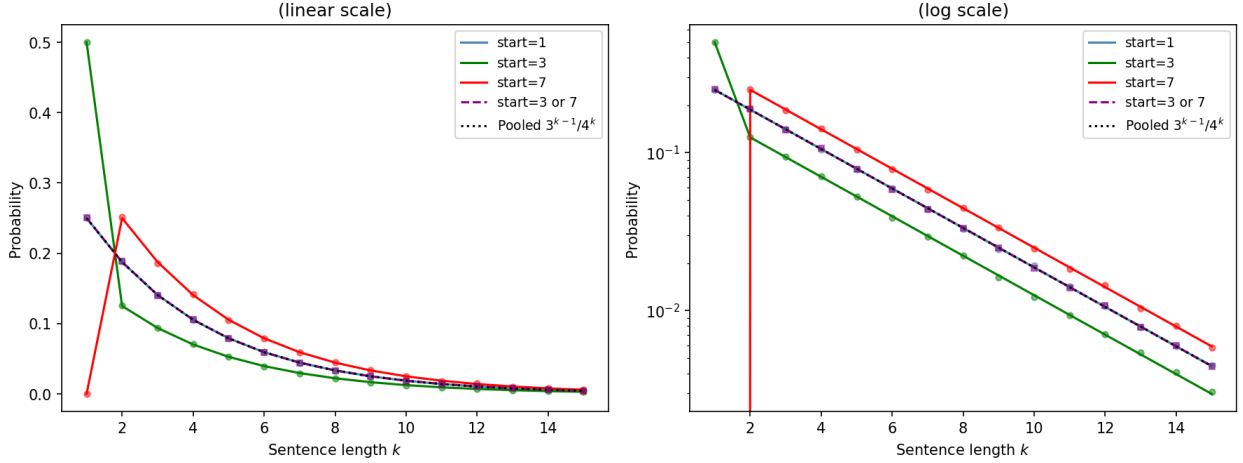


Figure 3: Steiner sentence length distribution stratified by start class $b_0 \bmod 8$: start=1 (steelblue), start=3 (green), start=7 (red), combined start=3 or 7 (purple dashed), pooled $3^{k-1}/4^k$ (black dotted); solid/dashed lines = exact per-class Markov-chain theory; dots/squares = empirical. Left: linear scale; right: log scale. Start=7 is shifted right (forbidden at $k = 1$); start=3 is shifted left ($P(k = 1) = 1/2$); they cancel exactly in the combined series.

Theorem 2.1 is not at fault: $P[7][3] = P[7][7] = 1/2$ as asymptotic densities over all $n \equiv 7 \pmod{8}$ is correct. The flaw is the implicit assumption that the mod-16 sub-classes of class 7 are equiprobable within each sentence, which is only true in the pooled ensemble (Proposition 3.2). Within a sentence starting at a specific $b_0 \equiv 7 \pmod{8}$, the mod-16 sub-class of b_0 is fixed, and subsequent class-7 nodes inherit that sub-class deterministically (Remark 2.4), so consecutive class-7 transitions are correlated rather than independent — giving slightly heavier tails than the pooled Markov model predicts.

Start=3 is unaffected because $P[3][1] = P[3][5] = 1/2$ is *exact* at mod 8: for $n \equiv 3 \pmod{8}$ the valuation $v_2(3n + 1) = 1$ is fixed, so each transition is a genuinely independent coin flip with no sub-modulus memory. The combined start=3 or 7 stratum inherits the start=7 tail excess at half weight, giving $p = 0.04$. The pooled fit is unaffected because the start=7 excess is diluted across all four start classes, and the asymptotic equiprobability holds in aggregate.

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